Computer Science & DA

Warehousing

Lecture -08

By- Vishvadeep Gothi sir

Recap of Previous Lecture

- **Topic** Data Discretization
- **Topic** Clustering
- **Topic** Data Generalization
- **Topic** Concept Hierarchy

Topics to be Covered

Topic Sampling

Topic Types of Sampling

Topic Compression

Topic : Sampling

The data-analysis practice of analyzing a subset of data in order to uncover meaningful information from a larger data set.

$$\text{standard deviation} = \sqrt{\frac{\Sigma(x_i - \mu)^2}{N}}$$

$$\text{S.D. of sample data} = \sqrt{\frac{\Sigma(x_i - \mu)^2}{n-1}}$$

n = Total no. of data items in sample

Topic : Sampling

Simple Random Sampling:

There is an equal probability of selecting any particular item.

Sampling without replacement:

As each item is selected, it is removed from the population.

Sampling with replacement:

Objects are not removed from the population as they are selected for the sample. In sampling with replacement, the same object can be picked up more than once

Stratified sampling:

Split the data into several partitions; then draw random samples from each partition

Topic : Sampling

8000 points 2000 Points 500 Points

Topic : Sampling

Benefits:

Time savings

Cost savings

Accuracy

Flexibility

Bias elimination

Topic : Sampling

Challenges:

Risk of bias

Determining the sample size

Sampling error

Topic : Data Sampling Methods

- **Probability Data Sampling:**
  - Uses random numbers that correspond to points in a dataset. This ensures that there is no correlation between sample selections.
  - Samples are chosen from a larger population to suit various statistical methods.
  - It is the best approach when the objective is to create a sample that accurately represents the population.

Topic : Data Sampling Methods

- **Non-Probability Data Sampling:**
  - The sample is identified and extracted based on researchers' or analysts' subjective judgment.
  - A limitation is that it can be subject to bias and may not accurately represent a larger population.
  - Used when expediency or simplicity are required.

Topic : Data Sampling Methods

- **Probability Data Sampling:**

1. Simple Random Sampling
2. Stratified Sampling
3. Cluster Sampling
4. Systematic Sampling or Systematic Clustering

- **Non-Probability Data Sampling:**

1. Convenience Sampling
2. Consecutive Sampling or Quota Sampling
3. Snowball Sampling or Referral Sampling
4. Purposive Sampling or Judgmental Sampling

Topic : Simple Random Sampling

Gives every element in the dataset an equal probability of being selected

Benefit:

Can be done easily and inexpensively, and it removes bias from the sampling process

Limitation:

A limitation of simple random sampling is unrepresentative groupings, because researchers do not have control over the sample selection

Topic : Stratified Sampling

- Split the data into several partitions; then draw random samples from each partition
- Used when researchers require groups based on a common factor.

- The process of classifying the population introduces bias.

Topic : Cluster Sampling

Divides datasets into subsets, or clusters, according to a defined factor. Then, samples are randomly selected as one of the clusters.

A more complicated variation of ~~customer~~ cluster sampling is multistage sampling. This method separates clusters based on different criteria then samples this subset. This can be done any number of times.

Topic : Systematic Sampling or Systematic Clustering

Creates samples by setting an interval at which to extract data from the dataset according to a predetermined rule.

every 3rd data ⟹ sample

Topic : Convenience Sampling

- Collects data from an easily-accessible and available group
  Example: asking people outside of a store to take a survey
- The collected elements are based on accessibility and availability
- Often used as a preliminary step and introduces significant bias

Topic : Consecutive Sampling or Quota Sampling

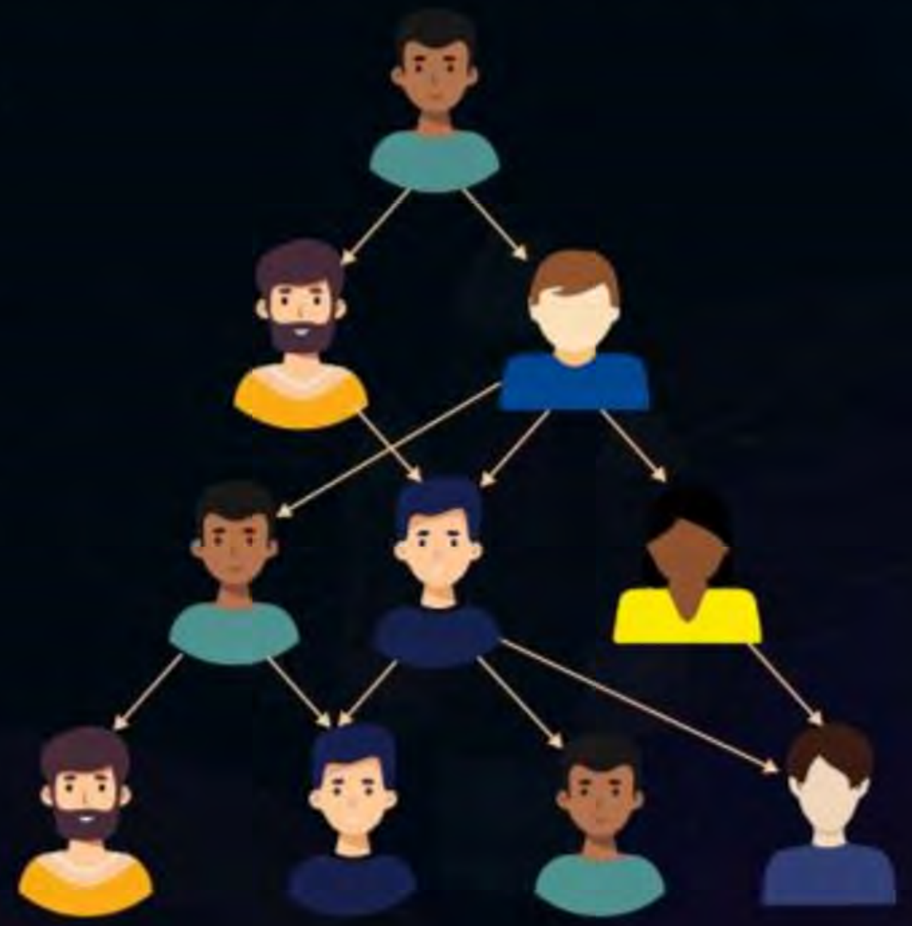
- Gathers data from sources that meet the criteria until the desired sample size is attained
- Ensures equal representation in subsets within the sample
- The proportion of the sample does not match the proportion of the group to the population

Topic : Snowball Sampling or Referral Sampling

Collects elements for the dataset by gathering those that are related, in essence getting help from the existing elements to refer to the others as samples that fit in the population

- Gathering a small group then asking them to invite friends to join

- Used when the population is unknown, and it is difficult to build a dataset.
- Based on referrals, hence the chance of bias is increased.



Topic : Purposive Sampling or Judgmental Sampling

Selects the sample based on predefined criteria established by the researcher according to the purpose of the research.

Topic : Common Data Sampling Errors

- It is the deviation between the estimate of an ideal sample and the actual population.

- Data sampling is meant to create samples that are a representative subset of the population, with the sample mean being equal to the mean of the population.

Topic : Data Sampling Bias

1. **Response bias**
   Occurs during data collection when a respondent or participant gives inaccurate or false answers to a question

2. **Voluntary response bias**
   Occurs when individuals are self-selected or choose to participate

3. **Non-response bias**
   Occurs when participants do not respond in whole or part

4. **Convenience bias**
   Occurs when a sample is taken from individuals that are conveniently available

Topic : Data Compression

- Data compression is the process of encoding, restructuring or otherwise modifying data in order to reduce its size.
- Fundamentally, it involves re-encoding information using fewer bits than the original representation.

Topic : Data Compression Trade off

- Compression and decompression can increase computational complexity and overhead.
- Lossy compression methods can lead to information loss, distortion, artifacts, and changes in the data's distribution or statistics.
- Compression may reduce interpretability or transparency of the data and affect the accuracy or validity of data mining results.

Topic : Data Compression

1. **Lossless:**

   Removes bits by locating and removing statistical redundancies. Because of this technique, no information is actually removed.

2. **Lossy:**

   Lowers size by deleting unnecessary information and reducing the complexity of existing information.

Topic : Lossless Data Compression

1. Lempel Ziv Welch Encoding
2. Huffman Encoding
3. RLE (Run Length Encoding)
4. Dictionary Coder (LZ77, LZ78, LZR, LZW, LZSS, LZMA, LZMA2)
5. Prediction by Partial Matching (PPM)
6. Deflate
7. Content Mixing
8. Adaptive Huffman Coding
9. Shannon Fano Encoding
10. Arithmetic Encoding
11. ZStandard
12. Bzip2 (Burrows and Wheeler)

Topic : Lossy Data Compression

1. Transform coding
2. Discrete Cosine Transform
3. Discrete Wavelet Transform
4. Fractal Compression

Topic : Lempel–Ziv–Welch Algorithm

LZW

Universal lossless data compression algorithm

- Parse the source sequence into phrases such that each phrase is the shortest phrase not seen earlier (incremental parsing)
- Describe(encode)each new phrase by describing the index of the phrase from the past that forms its prefix and the new symbol at the end of the phrase.

Co Conut

A B A B B A A B B

Topic : Lempel–Ziv–Welch Algorithm

Example:
VVDVDDDVDVVDVDDDVDDVDV

Total characters = 22 ⇒ each character = 8 bits
Hence total bits without compression to store all characters = 22*8 bits

= 176 bits

Topic : Lempel–Ziv–Welch Algorithm

Example:
VVDVDDDVDVVDVDDDVDDVDV

	V	VD	VDD	D	VDV	VDVD	DD	VDDV	DV
Position									
Sequence	1	2	3	4	5	6	7	8	9
Numerical Presentation	ϕV	1D	2D	ϕD	2V	5D	4D	3V	4V
Code	0	11	101	1	100	1011	1001	110	1000

$\phi \Rightarrow$ no prefix

2 unique characters

V D

V $\Rightarrow$ 0

D $\Rightarrow$ 1

final code must have each phrase represented by equal no. of bits. Hence whichever phase has max bits, make each phrase code equal to it.

Final code	1	2	3	4	5	6	7	8	9
	0000	0011	0101	0001	0100	1011	1001	0110	1000

Total no. of bits used = $9 * 4 = 36$ bits

no. of bits saved $= 176 - 36 = 140$ bits

⇒ Last most phrase may not be unseen.

Lempel–Ziv–Welch Algorithm

Homework

Example:

AABABABBABABBBAABAB

Position									
Sequence									
Numerical Presentation									
Code									

2 mins Summary

Topic Sampling

Topic Types of Sampling

Topic Compression

Happy Learning

THANK - YOU